MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG1DSCBOT100** Course Title: **Fascinating world of plant sciences**

Time: 1.5 Hrs **Maximum Marks: 50**

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part **A**:
Short answers
Answer 10 questions
Each question carries 1 mark

	Questions	Taxonomy	CO
1	Who is considered as the "Father of Botany"?	Remember	1
2	Which alkaloid is extracted from Quinine Tree bark?	Remember	1
3	Signify the role of scientific diagrams	Apply	3
4	Name the lichen species, that is used in traditional medicine and as a food additive?	Remember	2
5	List two examples of psychoactive plants consumed by animals.	Remember	2
6	Name an industry that could benefit from Citrus maxima-inspired designs	Remember	2
7	List any two peculiar adaptations of Insectivorous plants	Remember	2
8	Arctic poppy is an example of how smart plants are in adapting to challenging environments. Evaluate the statement.	Evaluate	2
9	What is the role of the rotating wheel in a rotary microtome?	Understand	3
10	A student is examining a leaf and wishes to understand the ultrastructure of chloroplasts. What is the best microscopy technique for this purpose?	Apply	3
11	Define Remote Sensing?	Remember	3
12	Name the thermostable enzyme used in PCR	Remember	3

Part **B**:
Short Essay Type Questions
Answer 6 Questions
Each question carries 5 marks

13	Pteridophytes are more advanced than bryophytes. Justify.	Analyse	2
14	Briefly analyse the evolutionary trends in land plants	Analyse	2
15	What are the advantages and limitations of free hand sectioning	Understand	3
16	Describe artificial system of classification of plants	Understand	3

17	Comment on the importance of botanical illustrations	Understand	3
18	Comment of relevance and scope of plant anatomical and plant physiological research in our life.	Understand	4
19	What are the different component parts of a GIS framework?	Understand	3
20	Describe how smart devices can be used in agriculture to manage water stress?	Apply	5

Part **C**:
Essay Type Questions
Answer any one question
Question carries 10 marks

21	Discuss electrophoresis, its principle, types, and application.	Understand	3
22	Compare and contrast gymnosperms and angiosperms	Analyse	2